



General Description

- Latest Trench Power AlphaMOS (αMOS LV) technology
- Very Low $R_{DS(ON)}$ at 2.5V V_{GS}
- Low Gate Charge
- ESD protection
- RoHS and Halogen-Free Compliant

Application

- DC/DC Converters

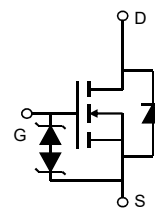
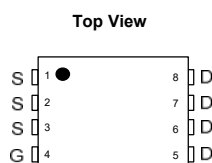
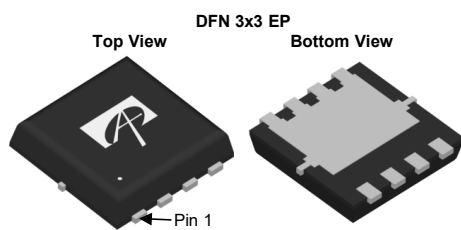
Product Summary

V_{DS}	30V
I_D (at $V_{GS}=10V$)	28A
$R_{DS(ON)}$ (at $V_{GS}=10V$)	< 3.3mΩ
$R_{DS(ON)}$ (at $V_{GS}=4.5V$)	< 4mΩ
$R_{DS(ON)}$ (at $V_{GS}=2.5V$)	< 5.8mΩ

Typical ESD protection

100% UIS Tested
100% R_g Tested

HBM Class 3B



Absolute Maximum Ratings $T_A=25^\circ\text{C}$ unless otherwise noted

Parameter	Symbol	Maximum	Units
Drain-Source Voltage	V_{DS}	30	V
Gate-Source Voltage	V_{GS}	± 12	V
Continuous Drain Current ^G	I_D	28	A
$T_C=25^\circ\text{C}$		22	
$T_C=100^\circ\text{C}$			
Pulsed Drain Current ^C	I_{DM}	112	
Continuous Drain Current	I_{DSM}	25	A
$T_A=25^\circ\text{C}$		20	
$T_A=70^\circ\text{C}$			
Avalanche Current ^C	I_{AS}	28	A
Avalanche energy $L=0.05\text{mH}$ ^C	E_{AS}	20	mJ
V_{DS} Spike	V_{SPIKE}	36	V
Power Dissipation ^B	P_D	32	W
$T_C=25^\circ\text{C}$		12.8	
$T_C=100^\circ\text{C}$			
Power Dissipation ^A	P_{DSM}	3.1	W
$T_A=25^\circ\text{C}$		2	
$T_A=70^\circ\text{C}$			
Junction and Storage Temperature Range	T_J, T_{STG}	-55 to 150	$^\circ\text{C}$

Thermal Characteristics

Parameter	Symbol	Typ	Max	Units
Maximum Junction-to-Ambient ^A	$R_{\theta JA}$	30	40	$^\circ\text{C/W}$
$t \leq 10\text{s}$				
Maximum Junction-to-Ambient ^{A D}	$R_{\theta JA}$	60	75	$^\circ\text{C/W}$
Steady-State				
Maximum Junction-to-Case	$R_{\theta JC}$	3.1	3.9	$^\circ\text{C/W}$
Steady-State				

Electrical Characteristics (T_J=25°C unless otherwise noted)

Symbol	Parameter	Conditions	Min	Typ	Max	Units
STATIC PARAMETERS						
BV _{DSS}	Drain-Source Breakdown Voltage	ID=250 A, V _{GS} =0V	30			V
I _{DSS}	Zero Gate Voltage Drain Current	V _{DS} =30V, V _{GS} =0V T _J =125°C			1 5	A
I _{GSS}	Gate-Body leakage current	V _{DS} =0V, V _{GS} =±10V			±10	A
V _{GS(th)}	Gate Threshold Voltage	V _{DS} =V _{GS} , I _D =250 A	0.4	0.8	1.2	V
R _{DS(on)}	Static Drain-Source On-Resistance	V _{GS} =10V, I _D =20A T _J =125°C		2.6 3.6	3.3 4.5	mΩ
		V _{GS} =4.5V, I _D =18A		3	4	mΩ
		V _{GS} =2.5V, I _D =16A		4.4	5.8	mΩ
g _{FS}	Forward Transconductance	V _{DS} =5V, I _D =20A		103		S
V _{SD}	Diode Forward Voltage	I _S =1A, V _{GS} =0V		0.62	1	V
I _S	Maximum Body-Diode Continuous Current ^G				28	A
DYNAMIC PARAMETERS						
C _{iss}	Input Capacitance	V _{GS} =0V, V _{DS} =15V, f=1MHz		2250		pF
C _{oss}	Output Capacitance			800		pF
C _{rss}	Reverse Transfer Capacitance			65		pF
R _g	Gate resistance	V _{GS} =0V, V _{DS} =0V, f=1MHz	1.5	3.0	4.5	Ω
SWITCHING PARAMETERS						
Q _g (10V)	Total Gate Charge	V _{GS} =10V, V _{DS} =15V, I _D =20A		37	50	nC
Q _g (4.5V)	Total Gate Charge			16	22	nC
Q _{gs}	Gate Source Charge			3.2		nC
Q _{gd}	Gate Drain Charge			5.2		nC
t _{D(on)}	Turn-On DelayTime	V _{GS} =10V, V _{DS} =15V, R _L =0.75Ω, R _{GEN} =3Ω		5		ns
t _r	Turn-On Rise Time			3		ns
t _{D(off)}	Turn-Off DelayTime			47.5		ns
t _f	Turn-Off Fall Time			11.3		ns
t _{rr}	Body Diode Reverse Recovery Time	I _F =20A, dI/dt=500A/ s		16		ns
Q _{rr}	Body Diode Reverse Recovery Charge	I _F =20A, dI/dt=500A/ s		23		nC

A. The value of R_{θJA} is measured with the device mounted on 1in² FR-4 board with 2oz. Copper, in a still air environment with T_A=25° C. The Power dissipation P_{DSM} is based on R_{θJA} ≤ 10s and the maximum allowed junction temperature of 150° C. The value in any given application depends on the user's specific board design.

B. The power dissipation P_D is based on T_{J(MAX)}=150° C, using junction-to-case thermal resistance, and is more useful in setting the upper dissipation limit for cases where additional heatsinking is used.

C. Single pulse width limited by junction temperature T_{J(MAX)}=150° C.

D. The R_{θJA} is the sum of the thermal impedance from junction to case R_{θJC} and case to ambient.

E. The static characteristics in Figures 1 to 6 are obtained using <300 s pulses, duty cycle 0.5% max.

F. These curves are based on the junction-to-case thermal impedance which is measured with the device mounted to a large heatsink, assuming a maximum junction temperature of T_{J(MAX)}=150° C. The SOA curve provides a single pulse rating.

G. The maximum current rating is package limited.

H. These tests are performed with the device mounted on 1 in² FR-4 board with 2oz. Copper, in a still air environment with T_A=25° C.

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TYPICAL ELECTRICAL AND THERMAL CHARACTERISTICS

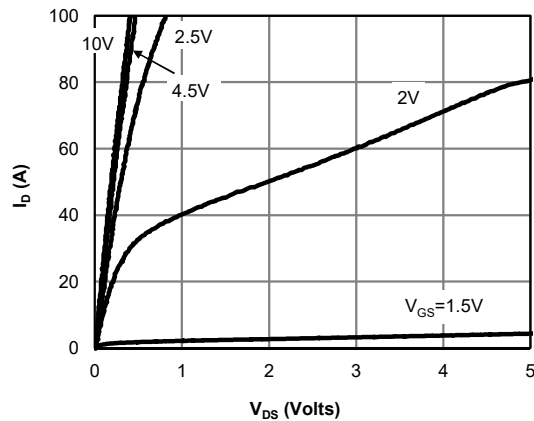


Fig 1: On-Region Characteristics (Note E)

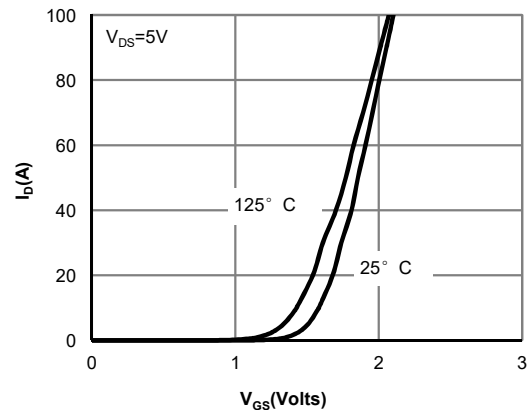


Figure 2: Transfer Characteristics (Note E)

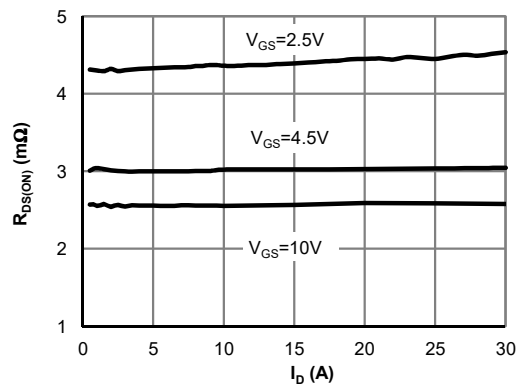


Figure 3: On-Resistance vs. Drain Current and Gate Voltage (Note E)

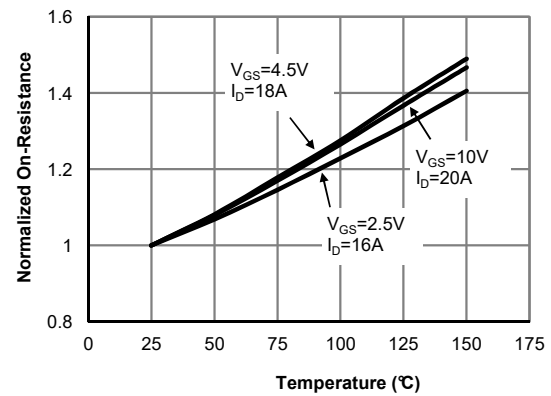


Figure 4: On-Resistance vs. Junction Temperature (Note E)

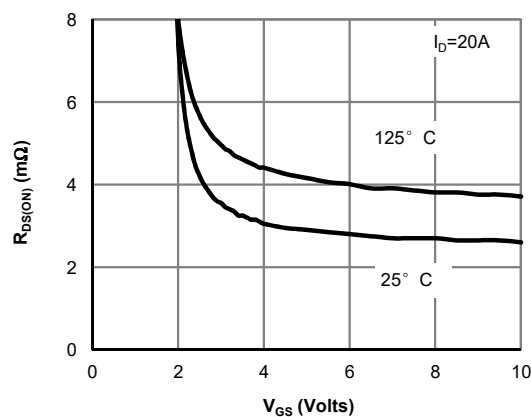


Figure 5: On-Resistance vs. Gate-Source Voltage (Note E)

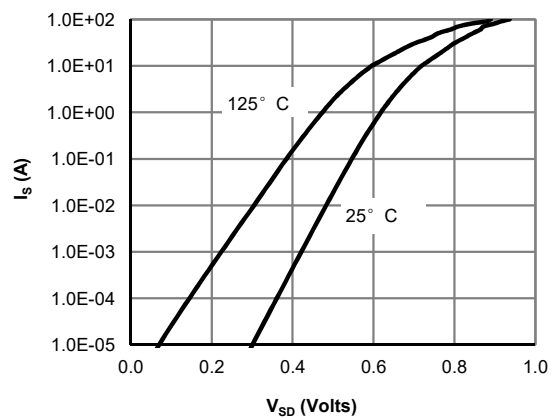
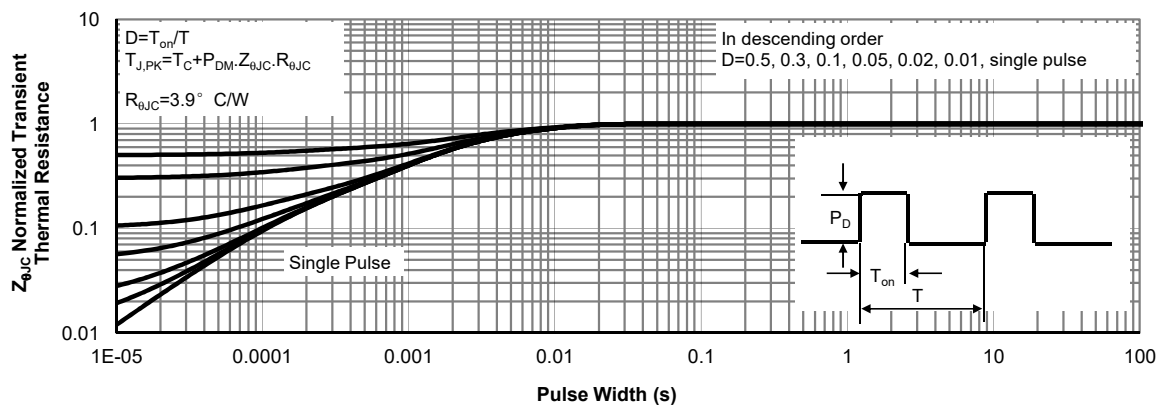
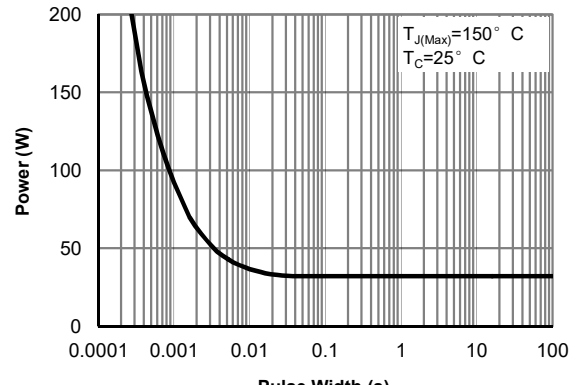
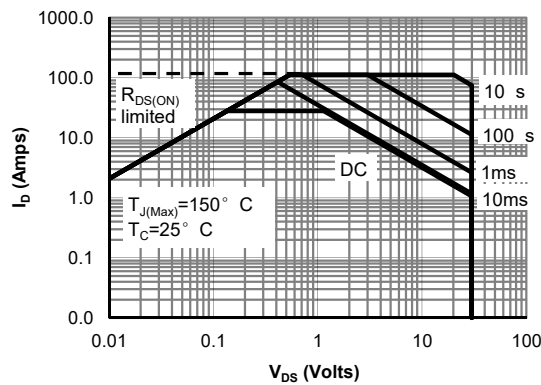
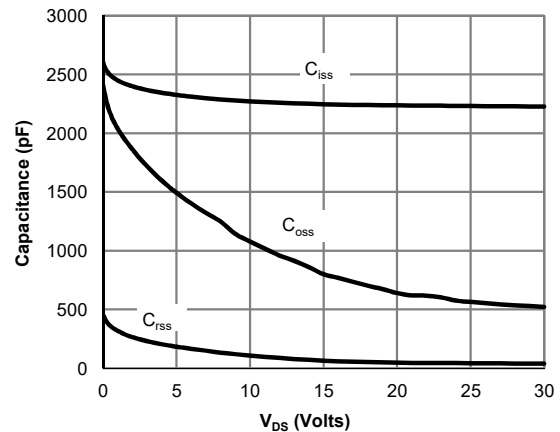
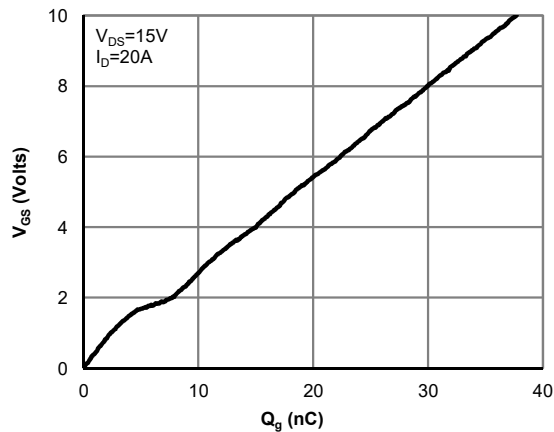


Figure 6: Body-Diode Characteristics (Note E)

TYPICAL ELECTRICAL AND THERMAL CHARACTERISTICS



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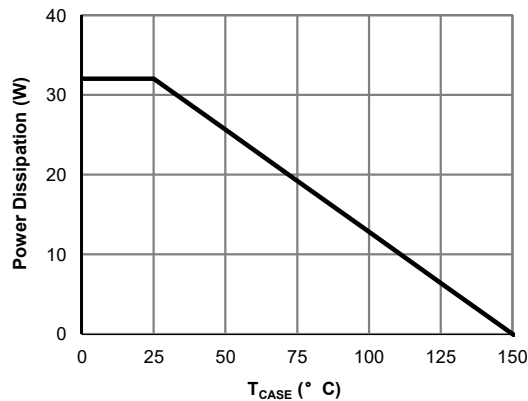


Figure 12: Power De-rating (Note F)

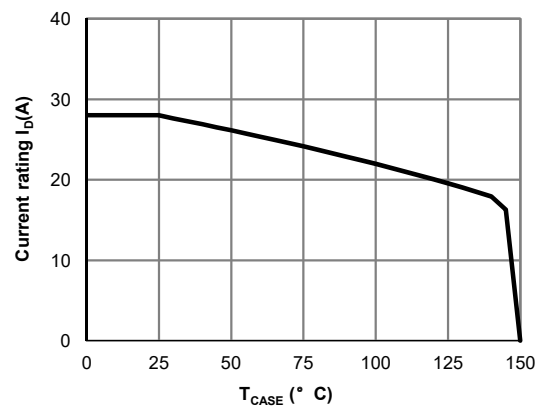


Figure 13: Current De-rating (Note F)

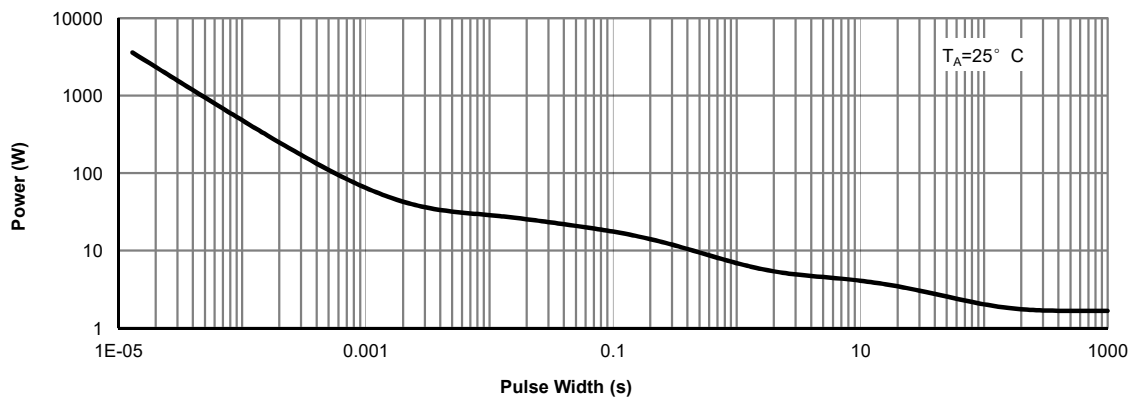


Figure 14: Single Pulse Power Rating Junction-to-Ambient (Note H)

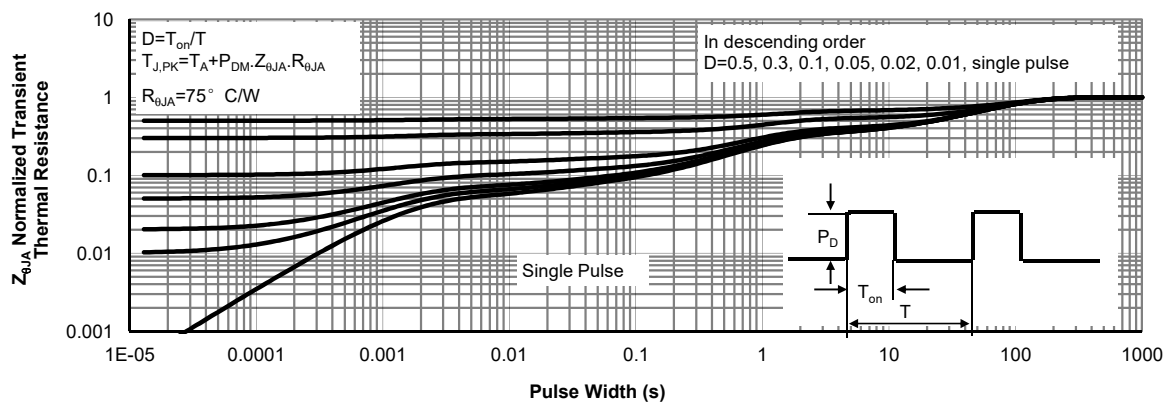
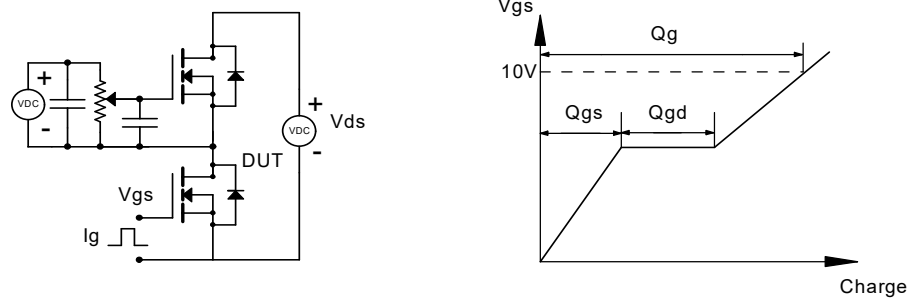
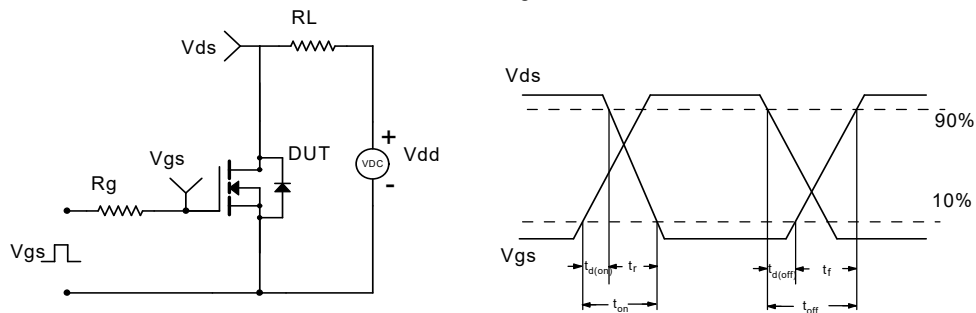


Figure 15: Normalized Maximum Transient Thermal Impedance (Note H)

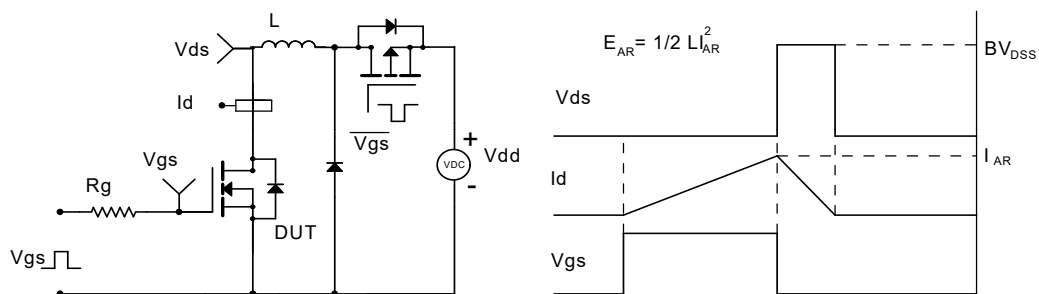
Gate Charge Test Circuit & Waveform



Resistive Switching Test Circuit & Waveforms



Unclamped Inductive Switching (UIS) Test Circuit & Waveforms



Diode Recovery Test Circuit & Waveforms

